PRACTICE QUESTIONS

Case Studies

Every downloadable case study from the LMS, summarised with its data and questions. Attempt them first; answers are in the Case Study Resolutions PDF.

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

Interactive version (every tab clickable):
https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

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PYTHON FOR DATA SCIENCE

Python Exercise 1: Basic Python Programming (20 questions)

Twenty short programs covering operators, input/output, loops, data types, list comprehensions, lambda/filter and user-defined functions. Where a question asks for console input, the solution uses a fixed sample value so the output is shown; replace it with input() in Jupyter.

Data

No data files.

Tool and module

Python (core)
Python for Data Science

Questions

1 Output of 5 + 4 * 9 % (3 + 1) / 6 - 1?

2 Check if a number is odd or even (number taken as input).

3 Print the multiplication table of a number.

4 Numbers between 2000 and 3200 divisible by 7 but not a multiple of 5 (comma-separated, one line).

5 Count elements of each data type in [2, 3, 'Py', '10', 1, 'SQL', 5.5, True, 3, 'John', None, 7].

6 Add all numeric values in the list.

7 Concatenate all strings with a hyphen.

8 UDF returning sum of numbers (excluding bool) and count of strings.

9 Get odd numbers from [5, 7, 22, 97, 54, 62, 77, 23, 73, 61] using (i) loop (ii) list comprehension (iii) lambda + filter.

10 UDF returning descriptives [sum, count, min, mean, max] for n numbers.

11 UDF for area of square/rectangle/circle: shape positional, dimensions as kwargs, with validation.

12 UDF to reconcile two month lists (matched, only in l1, only in l2).

13 UDF to check if a number is prime.

14 Factorial of given numbers (comma-separated output).

15 Dictionary {i: i*i} for 1..n.

16 Comma-separated numbers → list and tuple.

17 Sort comma-separated words alphabetically.

18 Remove duplicate words and sort.

19 Count upper-case and lower-case letters.

20 Reverse a string.

PYTHON FOR DATA SCIENCE

Python Exercise 2: Data Importing, Manipulation, Analysis and Visualisation (90 questions)

Ninety guided exercises in 8 sections: creating DataFrames, deleting data, filtering and sorting, aggregation, lambda/apply, joins, date-time operations and basic charts. Each answer below was run on the provided files.

Data

wine.txt, chipotle.tsv, occupation.csv, student-mat/por.csv, cars1/2.csv, wind.data, appl_1980_2014.csv, Online_Retail.csv, FMCG_Company_Data_2019.csv

Tool and module

Python (pandas, numpy, matplotlib)
Python for Data Science

Questions

1 Import the necessary libraries.

2–3 Create the dictionary and assign it to a DataFrame called pokemon.

4 Reorder columns to name, type, hp, evolution, pokedex.

5 Add a column 'place'.

6 Data type of each column.

7 Info of the DataFrame.

8 Import wine.txt without a header row.

9 Delete the 1st, 4th, 7th, 9th, 11th, 13th and 14th columns.

10 Name the columns alcohol, malic_acid, alcalinity_of_ash, magnesium, flavanoids, proanthocyanins, hue.

11 Set the first 3 values of alcohol to NaN.

12 Set rows 3 and 4 of magnesium to NaN.

13 Fill NaN with 10 in alcohol and 100 in magnesium.

14 Count missing values in all columns.

15 Array of 10 random numbers up to 10.

16 Set rows at those random positions to NaN in alcohol.

17 How many missing values now?

18 Print only non-null alcohol values.

19 Delete rows with missing values.

20 Reset the index.

21 chipotle: how many products cost more than $10?

22 Print the DataFrame and its info.

23 Price of each item: drop duplicates on item_name and quantity, keep item_name and item_price, sort high to low.

24 Sort by item name.

25 Quantity of the most expensive item ordered.

26 How many times was Veggie Salad Bowl ordered?

27 How many times did people order more than one Canned Soda?

28 Import occupation.csv as users.

29 Mean age per occupation.

30 Male ratio per occupation, sorted high to low (use np.where).

31 Min and max age per occupation.

32 Mean age per occupation and gender.

33 Percentage of women and men per occupation.

34 Import student-mat and student-por and append them as df.

35 Slice columns from school to guardian.

36 Lambda function that capitalises a string.

37 Capitalise Mjob and Fjob.

38 Print the last rows.

39 Why is the original still lowercase? Fix it.

40 Function majority → boolean column legal_drinker (age > 17).

41 Multiply every number in the dataset by 10.

42 Import cars1 and cars2.

43 Inspect cars1 (type, head, tail, columns, info, dtypes, shape, describe, value_counts...).

44 Remove the unnamed blank columns from cars1.

45 Number of observations in each dataset.

46 Join cars1 and cars2 into cars.

47 Random 'owners' Series between 15,000 and 73,000.

48 Add owners to cars.

49 Current date-time, year, month, week number, weekday, day of year, day of month, day of week.

50 Convert 'Jul 1 2014 2:43PM' to datetime.

51 Subtract five days from 2015-06-22.

52 Unix timestamp 1284105682 to readable date.

53 Convert ['07Sep59','01Jan55','15Dec47','11Jul42'] to dates in 19XX.

54 Days between two dates.

55 Convert '15Dec1989' and display as 'Friday, 15 Dec 89'.

56–58 Import wind.data, build a proper date index, fix years < 70 to 19XX, dtype datetime64.

59 Missing values per location.

60 Total non-missing values.

61 Mean wind speed over all locations and times.

62 loc_stats: min, max, mean, std per location.

63 day_stats: min, max, mean, std across locations per day.

64 Average January wind speed per location (all years).

65 Mean wind speed for each month (Jan 1961 and Jan 1962 different).

66 Weekly min/max/mean/std across locations for the first 52 weeks (weeks start 2 Jan 1961).

67 Import appl_1980_2014.csv as apple.

68 Column types.

69–70 Convert Date to datetime and set as index.

71 Any duplicate dates?

72 Sort so the oldest date is first.

73 Last business day of each month.

74 Days between the first and last date.

75 How many months of data?

76 Plot Adj Close with figure size 13.5 × 9 inches.

77 Import Online_Retail.csv as online_rt.

78–79 Bar chart of the top 10 countries by Quantity, excluding UK and negative quantities.

80 Scatter of Quantity vs UnitPrice per CustomerID for the top 3 countries by revenue.

81 Import FMCG_Company_Data_2019.csv as company_data.

82 Line chart of Total Revenue by month.

83 Line chart of Total Units: dotted red line, legend lower right.

84 Multi-line plot of units per product per month.

85 Bar chart for Soap (and save it).

86 Grouped bar chart for Soap, Shampo, ToothPaste per month.

87 Histogram of Total Revenue.

88 Pie chart of total 2019 units by product.

89 Soap and FaceWash in subplots.

90 Box plot of Total Profit.

Python Case Study 1: Customer Analysis for Retail

A retail store wants to analyse day-to-day transactions and keep track of its customers across locations, including their purchases and returns across product categories. Build a report with the calculated metrics, summaries and inferences listed below.

Data

Customer.csv (5,647 customers: id, DOB, gender, city code) · Transactions.csv (23,053 rows: qty, rate, tax, total amount, store type; negative values are returns) · prod_cat_info.csv (category and sub-category codes).

Tool and module

Python (pandas, matplotlib)
Python for Data Science

Questions

1 Merge Customers, Product Hierarchy and Transactions as Customer_Final, keeping every customer who has transacted.

2a Summary report: column names and their data types.

2b Summary report: top and bottom 10 observations.

2c Five-number summary (min, Q1, median, Q3, max) for continuous variables.

2d Frequency tables for all categorical variables.

3 Histograms for all continuous variables and frequency bars for categorical variables.

4a Time period of the available transaction data.

4b Count of transactions where the total amount was negative.

5 Which product categories are more popular among female vs male customers?

6 Which city code has the maximum customers, and what share of customers is from that city?

7 Which store type sells the maximum products by value and by quantity?

8 Total amount earned from Electronics and Clothing categories at Flagship stores.

9 Total amount earned from Male customers under the Electronics category.

10 How many customers have more than 10 unique transactions, after removing transactions with negative amounts?

11 For customers aged 25–35: (a) total spend on Electronics and Books; (b) total spend between 1 Jan 2014 and 1 Mar 2014.

Python Case Study 2: Exploratory Analysis of Credit Card Data

PSPD Bank operates in 50+ countries. The CEO wants to understand customer spend and repayment behaviour to evaluate bankruptcy, fraud and collections risk and to make proactive offers.

Data

Customer Acquisition (100 customers: age, city, product, per-transaction limit, segment) · Spend (1,500 transactions by type) · Repayment (1,523 payments).

Tool and module

Python (pandas, matplotlib)
Python for Data Science

Questions

1a Replace age below 18 with the mean age.

1b If spend is more than the customer's limit, replace it with 50% of that customer's limit.

1c If repayment is more than the limit, replace it with the limit.

2a How many distinct customers exist?

2b How many distinct spend categories exist?

2c Average monthly spend by customers.

2d Average monthly repayment by customers.

2e If the monthly interest rate is 2.9%, what is the bank's profit for each month? (Profit = interest on positive (repayment − spend).)

2f Top 5 product types (spend categories).

2g Which city has the maximum spend?

2h Which age group spends the most?

2i Top 10 customers by repayment.

3 City-wise spend on each product per year, with a chart.

4a Monthly comparison of total spend, city-wise.

4b Yearly spend on air tickets.

4c Monthly spend for each product: is there seasonality?

5 UDF: top 10 customers per city by repayment, for a chosen product (Gold/Silver/Platinum) and time period (yearly or monthly).

Python Case Study 3: EDA and Hypothesis Testing on Insurance Claims

An insurer wants a 360-degree view of claims and customers, to clean the data, profile claims by segment, and test business hypotheses about claim amounts with statistical tests.

Data

claims.csv (1,100 claims: cause, date, area, police report, type, amount with $ sign, total policy claims, fraud flag) · cust_demographics.csv (1,085 customers: gender, DOB, state, segment).

Tool and module

Python (pandas, scipy)
Python for Data Science

Questions

1 Combine claims and customer data into a 360-degree view.

2 Data audit: are any data types mismatched with their business meaning?

3 Convert claim_amount to numeric by removing the $ sign.

4 Create an alert flag (1/0) for injury claims not reported to police.

5 Keep the most recent claim per customer and drop duplicates on customer ID.

6 Impute missing values (mean for continuous, mode for categorical).

7 Calculate age in years and categorise: Children <18, Youth 18–30, Adult 30–60, Senior >60.

8 Average claim amount by customer segment.

9 Total claim amount by incident cause, for claims at least 20 days before 1 Oct 2018.

10 How many adults from TX, DE and AK claimed for driver-related causes?

11 Pie chart of total claim amount by gender and segment (show %).

12 Which gender claimed the most for driver-related issues?

13 Which age group had the most fraudulent claims? Show a bar chart.

14 Monthly trend of total claim amount (months in chronological order).

15 Average claim by gender and age group, faceted by fraudulent vs non-fraudulent.

16 Hypothesis: is the amount claimed by males and females similar?

17 Hypothesis: is there a relationship between age category and segment?

18 Hypothesis: has the current year shown a significant rise in claim amounts vs the 2016-17 average of $10,000?

19 Hypothesis: is there a difference in claim amounts between age groups?

20 Hypothesis: is there a relationship between total number of policy claims and claim amount?

Python Case Study 4: Visualisations on Pharma Sales Data

A leading pharmaceutical company wants a dynamic view of sales KPIs at national, region and territory level for 2015 vs 2016, to see progress and failures and speed up decisions. Reshape the data with groupby, stack/unstack and pivot as needed.

Data

SalesData.csv (3,709 rows: account, region, division, state, tier, month, Sales and Units for 2015 and 2016).

Tool and module

Python (pandas, matplotlib)
Python for Data Science

Questions

1 Compare sales by region for 2016 vs 2015 with a bar chart.

2 Contribution of each region to 2016 sales (pie chart).

3 Compare total sales of 2015 and 2016 by Region and Tier.

4 In the East region, which state declined in 2016 vs 2015?

5 In the High tier, which divisions sold fewer units in 2016 than 2015?

6 Create a Qtr column from Month (Jan–Mar Q1, Apr–Jun Q2, Jul–Sep Q3, Oct–Dec Q4).

7 Compare quarter-wise sales for 2015 and 2016 (bar plot).

8 Composition of 2016 quarterly sales by tier: four pie charts, one per quarter.

PYTHON FOR DATA SCIENCE

Basic Statistics Case Study: Hypothesis Testing (5 business problems)

Five short business problems, each answered with the right statistical test: Lending Club interest rates, pricing-expert quotes, a treatment-facility re-engineering programme, software job prioritisation, and a Film on the Rocks customer survey.

Data

LoansData.csv (2,500 loans) · Price_Quotes.csv (12 orders, two experts) · Treatment_Facility.csv (20 months) · Priority_Assessment.csv (642 jobs) · Films.csv (330 survey responses).

Tool and module

Python (scipy)

Python for Data Science

Questions

1a Lending Club: does interest rate vary with loan amount (less interest for higher loans)?

1b Lending Club: does loan length affect interest rate?

1c Lending Club: does interest rate vary by loan purpose?

1d Lending Club: is there a relationship between FICO score and home ownership?

2 Pricing experts: is there a difference between Mary's and Barry's average quotes?

3 Treatment facility: did re-engineering change the critical-incident rate (and turnover)?

4 Job priorities: are high-priority jobs completed faster than medium, and medium faster than low?

5 Film on the Rocks survey: overall satisfaction, its drivers, patron profile and best media outlet.

PYTHON FOR DATA SCIENCE

Python End-to-End Case Study: Marketing Analytics for an E-commerce Marketplace

A leading Indian online marketplace wants data-driven insight on business performance: customers, sellers, products, channels, payments and satisfaction. Clean the data, then answer the sample questions below and add your own analyses.

Data

Sep 2016–Oct 2018. Tables: CUSTOMERS, SELLERS, PRODUCTS, ORDERS, ORDER_ITEMS, ORDER_PAYMENTS, ORDER_REVIEW_RATINGS, GEO_LOCATION (~100k orders).

Tool and module

Python (pandas, matplotlib)
Python for Data Science

Questions

1a High-level metrics: revenue, quantity, products, categories, sellers, locations, payment methods.

1b New customers acquired every month.

1c Month-on-month customer retention.

1d Revenue from new vs existing customers by month.

1e Trends and seasonality: by month, weekday, hour, and payment method.

1f Popular products by month, seller, state and category (top 10 products overall shown).

1g Popular categories by state.

1h Top 10 most expensive products sorted by price.

2a Segment customers into groups by revenue generated.

2b Segment sellers into groups by revenue generated.

3 Cross-selling: top 10 category combinations bought together in the same order.

4 Payment behaviour: how do customers pay, and which channel is used most?

5a Top 10 and bottom 10 categories by average rating.

5c Average rating by state and by month; link to delivery delay.

EXCEL (DATA VISUALIZATION & ANALYTICS)

Excel Case Study: Sportsmen Roster (Data Cleaning, Analysis, Report)

XYZ Co organises international sports events. Countries nominate sportsmen, and the team must standardise the membership roster (Stage 1), summarise it (Stage 2) and build a filterable report (Stage 3).

Data

Sheets: SPORTSMEN (50 members, 19 columns), SPORT (sport → indoor/outdoor), LOCATION (country code → country name and language, laid out horizontally).

Tool and module

Microsoft Excel
Data Visualization & Analytics (Excel)

Questions

1.1 Populate FULL NAME as PREFIX FIRSTNAME LASTNAME in UPPER CASE (column B).

1.2 Get COUNTRY NAME from the LOCATION sheet (column K).

1.3 Populate LANGUAGE from the LOCATION sheet (column L).

1.4 EMAIL: lastname.firstname@xyz.org for English speakers, @xyz.com for others, all lower case (column M).

1.5 Populate SPORT LOCATION (indoor/outdoor) from the SPORT sheet (column Q).

2.1 Show MEMBER ID as a 3-digit number (001, 002 ...).

2.2 Format BIRTHDATE as dd mmm' yyyy (e.g. 09 May' 1986).

2.3 Show WEIGHT with units (e.g. 80 kg).

2.4 SALARY in thousands: 2 decimals below 100,000, 1 decimal otherwise (87670 → 87.67 k; 123250 → 123.2 k).

Stage 2 Pivot (ANALYSIS!B3): COUNTRY in rows, GENDER in columns, count of members, no grand totals; plus the same cross-table with functions at G4.

Stage 3 Pivot report (REPORT!A3) with member details in tabular form, years only for birth date, no +/- buttons or grand totals, filter by SPORT LOCATION.

EXCEL (DATA VISUALIZATION & ANALYTICS)

Excel Case Study: Consumer Complaint Analysis

Analyse US consumer financial complaints: map state names from JSON, fix dates, report at company level, find the top companies and issues, and build a year-level KPI dashboard with priorities.

Data

Consumer_Complaints (one row per complaint: company, product, issue, state, channel, received/resolved dates, timely response, disputed) · State_Code_Name (JSON text per state).

Tool and module

Microsoft Excel
Data Visualization & Analytics (Excel)

Questions

T1.1–1.2 Split the JSON text in State_Code_Name into STATE and CODE columns.

T1.3–1.5 Add State_Name to Consumer_Complaints by lookup, and count the #N/A records.

T1.6–1.7 Complaints by State code where State_Name did not map, and why.

T2 Resolution time in days, YEAR column, and US-FY quarter (Q1 = Jan–Mar).

T3 Company-level report: complaints (desc), number and % not timely, number and % disputed, average delay.

T4 Top 5 companies and top 5 issues by complaints; monthly trend chart.

T5 Year-level KPIs with YoY change; product and channel mix; priority view.

SQL (DATA VISUALIZATION & ANALYTICS)

SQL Case Study (Basic): Retail Data Analysis

A retail store wants to understand customer behaviour from its point-of-sale data. Write one SQL query per question, assuming read-only access.

Data

Customer (5,647 rows) · Transactions (23,053 rows; negative Qty/amount = return) · prod_cat_info (23 rows). Join Transactions to prod_cat_info on BOTH prod_cat_code and prod_subcat_code.

Tool and module

SQL Server (T-SQL)
Data Visualization & Analytics (SQL)

Questions

DP1 Total number of rows in each of the 3 tables.

DP2 Total number of transactions that have a return.

DP3 Convert the date columns to valid date formats.

DP4 Time range of the transaction data, in days, months and years (separate columns).

DP5 Which product category does the sub-category 'DIY' belong to?

DA1 Which channel (store type) is most frequently used for transactions?

DA2 Count of male and female customers.

DA3 City with the maximum number of customers, and how many.

DA4 How many sub-categories are there under Books?

DA5 Maximum quantity of products ever ordered.

DA6 Net total revenue in Electronics and Books.

DA7 How many customers have more than 10 transactions, excluding returns?

DA8 Combined revenue from Electronics and Clothing at Flagship stores.

DA9 Total revenue from male customers in Electronics, by sub-category.

DA10 Percentage of sales and returns by sub-category; top 5 sub-categories by sales.

DA11 Net revenue from customers aged 25–35 in the last 30 days of transactions (from the max transaction date).

DA12 Product category with the maximum value of returns in the last 3 months of transactions.

DA13 Store type selling the maximum products by value and by quantity.

DA14 Categories whose average revenue is above the overall average.

DA15 Average and total revenue by sub-category, for the top 5 categories by quantity sold.

SQL (DATA VISUALIZATION & ANALYTICS)

SQL Case Study (Advanced): Mobile Manufacturer Data Analysis

The database 'Cellphones Information' holds cell phone sales. Create the schema from the creation script provided, then answer 10 questions. Each answer must return a single table and goes between the --Qn--BEGIN and --Qn--END marks of the answer template.

Data

Star schema: DIM_MANUFACTURER, DIM_MODEL, DIM_CUSTOMER, DIM_LOCATION, DIM_DATE and FACT_TRANSACTIONS (IDModel, IDCustomer, IDLocation, Date, TotalPrice, Quantity).

Tool and module

SQL Server (T-SQL)
Data Visualization & Analytics (SQL)

Questions

1 List all states with customers who bought cellphones from 2005 till today.

2 Which US state buys the most Samsung phones?

3 Number of transactions for each model per zip code per state.

4 The cheapest cellphone (with its price).

5 Average price of each model in the top 5 manufacturers by sales quantity, ordered by average price.

6 Customers and their average spend in 2009, where the average is above 500.

7 Any model in the top 5 by quantity in 2008, 2009 and 2010 simultaneously?

8 Manufacturer with the 2nd highest sales in 2009 and in 2010.

9 Manufacturers that sold in 2010 but not in 2009.

10 Top 100 customers: average spend and quantity by year, and % change in spend.

TABLEAU

Tableau Case Study: Laptop Sales Strategy (London retail chain)

A London retail chain with 15+ stores sells laptops and accessories. It wants a product strategy and pricing policy that maximises projected 2016 revenue, based on 2015 point-of-sale data.

Data

POS.csv (2015 sales: date, configuration, customer OS X/Y, store postcode, retail price) · Laptops.csv (864 configurations: screen, battery, RAM, CPU, HDD) · Store_Locations.csv (store OS X/Y) · London_postal_codes.csv.

Tool and module

Tableau

Tableau

Questions

1 Top 20 configurations sold in 2015 (make '20' dynamic).

2 Has the average price of these top 20 varied over time?

3 Are average prices of the top 20 consistent across stores?

4 Do sales vary by month or by day of month?

5 Do stores with lower average prices sell more?

6 Group the 864 configurations into 32 groups (High/Low for 5 parameters) and compare average price.

7 On average, how far do customers travel to buy laptops (overall and per store)?

8 Does store proximity help increase sales?

9–10 Store-view dashboard with interactive store filter, then a Story.

TABLEAU

Tableau Case Study: Sports Equipment Retailer

A global sports-equipment retailer's product manager wants to analyse revenue across retailer countries, products and time. Submit a .twbx with one worksheet per question.

Data

Sales Data (16,210 orders, 2012–2014: retailer country/type, order mode, product category/type/product, revenue, quantity, gross margin).

Tool and module

Tableau

Tableau

Questions

1 Market share of revenue by retailer country, with labels and no legend.

2 Revenue vs quantity by product: is there a correlation?

3 Quarterly revenue across the three years for each country.

4 Dual-axis chart of revenue and quantity by product type.

5 Set of top 5 product types by quantity: what happens to their revenue?

6 Total revenue and profit by retailer country (Profit = Revenue × Gross margin).

7–8 Top 15 products sold in 2014, with a dynamic year and a dynamic Top N.

9 Pareto chart of quantity by product type; colour the types making up 80% of quantity.

10 Year-on-year % growth by product category.

11 3-month moving average of revenue vs actual by month.

12 View showing total by country for a user-selected measure (Revenue, Profit or Quantity).